

## LAU FAU SHAN, Hong Kong

WMO: 450350

Lat: 22.469N

Lon: 113.984E

Elev: 117

StdP: 14.63

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 45.1       | 47.5 | 24.5                          | 18.7 | 53.7 | 29.7 | 23.9 | 54.1 | 19.5                  | 53.4 | 17.8 | 53.5 | 10.4                  | 30   | 0.377 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.6                    | 92.1            | 80.6 | 90.8 | 80.3 | 89.7 | 80.1 | 82.7                | 88.9 | 81.9 | 87.9 | 81.3 | 87.1 | 11.0                 | 250  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 81.0                            | 161.8 | 86.4 | 80.3 | 157.9 | 85.6 | 79.7 | 154.9 | 85.0 | 46.9          | 89.2 | 46.0 | 88.1 | 45.3 | 87.6 | 90.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 19.8          | 17.2 | 15.2 | 41.9                       | 95.5 | 4.6                | 1.1 | 38.6                                               | 96.3 | 36.0       | 97.0 | 33.4       | 97.6 | 30.1       | 98.4 |    |    |
| (5)               |      |      | 35.3                       | 86.3 | 3.6                | 2.6 | 32.7                                               | 88.2 | 30.7       | 89.7 | 28.6       | 91.2 | 26.0       | 93.0 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 74.2  | 60.2  | 63.2  | 67.4  | 73.8  | 79.7  | 82.8  | 84.0  | 83.6  | 82.4  | 78.3  | 71.5  | 63.2 | (6)  |
| (7)  |                                                                                  | DBStd       | 9.79  | 6.14  | 7.73  | 6.15  | 5.20  | 3.94  | 3.11  | 2.55  | 2.62  | 2.82  | 3.50  | 5.58  | 5.91 | (7)  |
| (8)  |                                                                                  | HDD50       | 12    | 6     | 5     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1    | (8)  |
| (9)  |                                                                                  | HDD65       | 456   | 170   | 114   | 47    | 7     | 0     | 0     | 0     | 0     | 0     | 0     | 14    | 104  | (9)  |
| (10) |                                                                                  | CDD50       | 8856  | 322   | 374   | 541   | 713   | 921   | 985   | 1055  | 1040  | 973   | 876   | 646   | 410  | (10) |
| (11) |                                                                                  | CDD65       | 3822  | 21    | 62    | 122   | 270   | 456   | 535   | 590   | 575   | 523   | 411   | 210   | 47   | (11) |
| (12) |                                                                                  | CDH74       | 35234 | 23    | 137   | 361   | 1475  | 4069  | 5920  | 6952  | 6639  | 5599  | 3075  | 909   | 75   | (12) |
| (13) |                                                                                  | CDH80       | 11675 | 1     | 11    | 27    | 238   | 1203  | 2133  | 2764  | 2515  | 1924  | 726   | 130   | 3    | (13) |
| (14) | Wind                                                                             | WSAvg       | 7.8   | 7.4   | 7.6   | 7.7   | 8.1   | 8.0   | 8.4   | 8.3   | 7.4   | 7.7   | 7.6   | 7.7   | 7.9  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 71.1  | 1.1   | 1.8   | 2.9   | 6.5   | 9.5   | 11.9  | 11.8  | 12.3  | 7.4   | 2.9   | 1.3   | 1.1  | (15) |
| (16) |                                                                                  | PrecMax     | 117.6 | 4.1   | 9.0   | 21.5  | 14.8  | 28.4  | 30.2  | 23.3  | 34.6  | 19.3  | 24.2  | 6.1   | 7.0  | (16) |
| (17) |                                                                                  | PrecMin     | 37.3  | 0.0   | 0.0   | 0.0   | 0.1   | 0.3   | 0.0   | 0.1   | 2.0   | 0.3   | 0.0   | 0.0   | 0.0  | (17) |
| (18) |                                                                                  | PrecStd     | 19.6  | 1.1   | 2.5   | 4.3   | 4.0   | 7.9   | 6.4   | 6.5   | 6.9   | 5.1   | 4.9   | 1.7   | 1.8  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 76.7  | 81.7  | 82.8  | 86.4  | 91.6  | 92.9  | 94.0  | 93.5  | 93.0  | 89.8  | 86.4  | 79.4 | (19) |
| (20) |                                                                                  |             | MCWB  | 67.3  | 71.3  | 72.1  | 76.6  | 79.4  | 80.7  | 81.5  | 81.2  | 79.5  | 76.6  | 74.0  | 68.9 | (20) |
| (21) |                                                                                  | 2%          | DB    | 73.1  | 77.9  | 80.2  | 84.4  | 89.2  | 90.9  | 91.7  | 91.6  | 90.9  | 87.5  | 83.2  | 76.0 | (21) |
| (22) |                                                                                  |             | MCWB  | 65.4  | 69.6  | 71.0  | 75.8  | 79.1  | 80.4  | 81.2  | 80.7  | 79.2  | 76.1  | 72.8  | 67.0 | (22) |
| (23) |                                                                                  | 5%          | DB    | 70.6  | 75.1  | 77.8  | 82.7  | 87.5  | 89.4  | 90.4  | 90.1  | 89.2  | 85.5  | 81.2  | 73.5 | (23) |
| (24) |                                                                                  |             | MCWB  | 63.5  | 68.4  | 70.3  | 75.0  | 78.5  | 80.4  | 80.8  | 80.6  | 79.2  | 74.6  | 71.8  | 65.0 | (24) |
| (25) |                                                                                  | 10%         | DB    | 68.3  | 72.7  | 75.7  | 81.0  | 85.7  | 87.8  | 89.0  | 88.7  | 87.7  | 83.8  | 79.0  | 71.1 | (25) |
| (26) |                                                                                  |             | MCWB  | 62.0  | 66.9  | 69.4  | 74.2  | 77.6  | 80.0  | 80.3  | 80.3  | 78.5  | 73.5  | 70.7  | 63.5 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 69.1  | 73.1  | 74.5  | 78.3  | 81.7  | 83.4  | 83.5  | 83.8  | 83.3  | 80.8  | 76.5  | 71.6 | (27) |
| (28) |                                                                                  |             | MCD B | 74.1  | 78.7  | 79.7  | 83.2  | 88.0  | 89.8  | 90.5  | 90.2  | 88.2  | 85.7  | 82.3  | 77.3 | (28) |
| (29) |                                                                                  | 2%          | WB    | 66.9  | 71.5  | 73.1  | 77.3  | 80.7  | 81.9  | 82.4  | 82.6  | 81.6  | 78.6  | 75.2  | 69.1 | (29) |
| (30) |                                                                                  |             | MCD B | 71.4  | 76.6  | 78.0  | 82.0  | 86.6  | 87.9  | 89.0  | 88.6  | 87.4  | 83.2  | 80.8  | 73.4 | (30) |
| (31) |                                                                                  | 5%          | WB    | 65.0  | 69.6  | 72.0  | 76.4  | 79.9  | 81.3  | 81.6  | 81.7  | 80.7  | 77.2  | 74.0  | 67.1 | (31) |
| (32) |                                                                                  |             | MCD B | 69.1  | 73.9  | 76.5  | 81.0  | 85.4  | 87.0  | 87.9  | 87.5  | 86.5  | 82.0  | 79.0  | 71.9 | (32) |
| (33) |                                                                                  | 10%         | WB    | 63.3  | 67.7  | 70.7  | 75.3  | 79.1  | 80.7  | 80.9  | 81.0  | 79.8  | 75.9  | 72.6  | 64.9 | (33) |
| (34) |                                                                                  |             | MCD B | 67.2  | 71.8  | 74.9  | 80.0  | 84.2  | 86.1  | 86.9  | 86.6  | 85.3  | 81.4  | 77.3  | 70.2 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 11.0  | 10.8  | 11.1  | 10.6  | 10.1  | 9.1   | 9.6   | 9.5   | 10.2  | 11.0  | 11.5  | 11.4 | (35) |
| (36) |                                                                                  |             | MCDBR | 13.0  | 13.5  | 13.2  | 12.0  | 11.8  | 10.7  | 11.1  | 11.4  | 12.1  | 12.3  | 12.9  | 13.4 | (36) |
| (37) |                                                                                  |             | MCWBR | 6.9   | 6.5   | 6.3   | 5.6   | 4.8   | 4.4   | 4.5   | 4.9   | 5.1   | 5.2   | 5.7   | 6.9  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 11.6  | 11.9  | 11.4  | 10.8  | 10.1  | 9.7   | 10.4  | 10.4  | 10.6  | 10.4  | 11.1  | 12.0 | (38) |
| (39) |                                                                                  |             | MCWBR | 6.7   | 6.3   | 6.2   | 5.7   | 4.7   | 4.7   | 4.8   | 5.2   | 5.1   | 5.1   | 6.4   | 7.1  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.587 | 0.639 | 0.767 | 0.745 | 0.586 | 0.552 | 0.538 | 0.606 | 0.641 | 0.688 | 0.582 | 0.573 | (40) |      |
| (41) |                                                                                  | taud        | 1.794 | 1.724 | 1.515 | 1.595 | 1.967 | 2.074 | 2.106 | 1.904 | 1.803 | 1.662 | 1.868 | 1.849 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 213   | 213   | 193   | 202   | 236   | 242   | 245   | 230   | 218   | 198   | 213   | 211   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 63    | 72    | 92    | 86    | 59    | 53    | 51    | 62    | 68    | 75    | 58    | 58    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 975   | 992   | 1059  | 1166  | 1406  | 1447  | 1689  | 1582  | 1486  | 1387  | 1129  | 1023  | (44) |      |
| (45) |                                                                                  | RadStd      | 160   | 155   | 144   | 185   | 185   | 214   | 180   | 140   | 108   | 139   | 129   | 152   | 45)  |      |

## Historical Trends

|                              |       | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) Station Only            | DBAvg | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (47) Regional (13 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page